

YUANJUN (JOHNNY) DAI

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EDUCATION

Case Western Reserve University Ph.D. Candidate in Computer Science · Advisor: Prof. An Wang (NSF CAREER Award) Research focus: in-kernel system observability, GPU telemetry for AI/ML workloads, and distributed system monitoring	08/2019 – Dec. 2026 (exp.) Cleveland, OH
University of Pittsburgh M.S. in Electrical and Computer Engineering Research focus: Computer Architecture and Microarchitecture	2014 Pittsburgh, PA
Northeastern University B.S. in Electrical Engineering	2008 Shenyang, China

SKILLS

Observability	eBPF (Kernel Tracing), Prometheus, Grafana, DCGM Exporter, Intel RAPL
Languages	Python, C/C++
Agentic AI	LangGraph, Multi-Agent Orchestration, Multi-Model Routing, Prompt Engineering
ML Frameworks	PyTorch (DDP/FSDP), vLLM, Megatron-LM, NCCL, BytePS, scikit-learn
Parallelism Paradigms	Data Parallel (DDP/FSDP), Tensor Parallel (TP), Pipeline Parallel (PP), Hybrid Parallel
Infrastructure	Kubernetes, Docker, CI/CD Pipelines (GitHub Actions), ArgoCD, Slurm
Systems & APIs	FastAPI, REST API, gRPC
Cloud & HPC	Lambda Labs HPC, Ohio Supercomputer Center, NSF FABRIC, CloudLab
Networking	SDN (Software-Defined Networking), P4, NPU/ASIC (Broadcom BCM SDK)

WORK EXPERIENCE

Software Engineer II Cisco Systems, Inc. Developed NPU/ASIC features for a 100 Gbps core router (Azure, AWS, Alibaba Cloud) using Broadcom BCM SDK; contributed to packet forwarding modules, reducing latency under heavy load.	06/2015 – 08/2016
Software Engineer Intern Broadcom (Emulex) Re-architected RTOS data acquisition (DMA + circular-buffer, 30% CPU reduction); ported BladeEngine ASIC CLI from Linux to Windows Server.	10/2014 – 05/2015
Technical Product Manager Xiaomi - Defined IoT appliance requirements via competitive analysis and user research; translated market needs into actionable engineering specifications. - Assessed technical feasibility and delivered architecture proposals covering trade-offs and risks to align engineering and business stakeholders.	07/2018 – 01/2019
Product Manager Midea Group Led technical PM for Fun Oven II across ML model training, Docker deployment on Alibaba Cloud, mobile app, and culinary algorithm design; won Red Dot Design Award, debuted at 2018 AWE Show · Red Dot ↗ · PR Newswire ↗	09/2016 – 06/2018

RESEARCH EXPERIENCE

Research Assistant Case Western Reserve University In-Kernel eBPF Sketch Systems for DML Network and System Resource Monitoring Paper (PSketch) / Code · Paper (eHashPipe) / Code PSketch: Designed a three-tier in-kernel network monitoring system in eBPF that classifies flows by behavior — priority flows get exact per-packet tracking via a hashmap, elephant flows are identified and tracked via a heavy flow table with voting-based kick-out collision resolution, and remaining mice flows are approximated via a 3-layer Count-Min Sketch; achieves O(n) Top-K detection, 96% accuracy, and <1.1% throughput overhead at 10 Gbps	08/2019 – Dec. 2026 (exp.)
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- **eHashPipe**: Designed a dual-pipeline in-kernel observability framework in **eBPF** for DML workloads — one pipeline profiles memory usage via HashPipe with cascading slot eviction and deallocation tracking, the other tracks per-thread CPU time by hooking the **sched_switch** tracepoint; delivers **10–14× finer temporal resolution** than top/htop at ~11ms latency

Scalable AI Model Serving Infrastructure on Kubernetes | [Code](#) / [Video](#)

- Designed a self-hosted LLM inference platform on **Kubernetes** (NGINX → **FastAPI** → **vLLM**) with **GitOps CI/CD** (**ArgoCD**), **GPU time-slicing**, hardware **MIG** partitioning, and custom-metrics **HPA** autoscaling
- Built 4-layer observability via **Prometheus** + **Grafana**: GPU hardware (**DCGM**: utilization, VRAM, temperature, power), **vLLM** inference (queue depth, KV cache, TTFT), gateway/router request counters and latency histograms per endpoint (**prometheus_client**), and K8s cluster metrics (**kube-prometheus-stack**)

DYNAMIX – Reinforcement Learning-Based Adaptive Batch Scheduling for Distributed ML | [Paper](#) / [Code](#)

- Designed a **PPO**-based **Reinforcement Learning (RL)** scheduler monitoring hardware signals (CPU/memory, bandwidth, retransmission rate via **eBPF** probes) and statistical signals (loss convergence, generalization) to dynamically adapt batch size, evaluated on **PyTorch DDP** and **Parameter Server (BytePS)**
- Achieves **46% wall-clock speedup** and **~6% accuracy gain** across **8–64 A100 GPUs** over fixed batch size baseline

PRISM – Priority-Based Selective Reliability for Distributed ML Training | [Paper](#)

- Designed a transport-layer library (**NCCL/Gloo** → **UDP**) hooking into **PyTorch DDP**'s AllReduce at bucket-ready events; classifies gradients by magnitude into high-priority (full **SACK**-based retransmission) and low-priority (local compensation via cached values)
- Supports kernel-bypass (**LibVMA**) and kernel-based (**GSO/GRO**) variants; achieves **8–15% throughput gain** for **CNNs** and **15–21% for LLMs** over TCP, matching NDP latency at 144-worker scale

Scotch – Elastic Distributed Overlay for Control-Plane Scalability and Load Balancing | [Paper](#) / [Code](#)

- Designed an **OvS**-based elastic overlay: when physical switch OFA is saturated, tunnels new flows to a **vSwitch** pool via **event-driven** Packet-In Edge messages; built a centralized **Ryu SDN** controller for **flow monitoring** and **load balancing** across vSwitches
- Separates small disruptive flows (terminated at vSwitches) from large flows (migrated to physical paths); achieves **>70% controller congestion reduction** on 34-node GENI Clos topology without hardware modification

AI-Powered Job Hunting Agent | [Code](#) / [Report](#)

- Built a **multi-agent AI platform** using **LangGraph** for automated job discovery and resume optimization — orchestrates specialized agents for collection, scoring, and refinement across 5 platforms, with a **6-dimension LLM matching framework** designed via **prompt engineering**
- Implemented **capability-aware multi-model routing** (**DeepSeek** for high-volume scoring, **Claude Opus** for quality-critical tailoring) and a **FastAPI** backend with cross-platform deduplication and health monitoring endpoints

LLM Fine-tuning for Multi-label Text Classification on Imbalanced Data | [Paper](#)

- Built a multi-label text classifier across 115 highly imbalanced labels — applied **inverse-frequency weighted loss** and **per-label threshold calibration** to improve long-tail performance
- Fine-tuned **PubMedBERT** via **PyTorch DDP** on 2×RTX A6000 GPUs; fine-tuned **Qwen-7B** via **NeMo + Megatron-LM** with **FSDP (ZeRO-3)** and **FlashAttention** to fit under GPU memory constraints; used **few-shot prompting** to mitigate hallucination in label generation

Side-Channel Attack Analysis and Behavioral Profiling of Distributed ML Systems | [Paper](#) / [Code](#)

- **Analyzed DML computation/communication behavior** in **MXNet BSP** Parameter Server: instrumented push/pull primitives and modified MXNet source code to capture **DML network behavior** and ground-truth per-layer traffic patterns
- Profiled **ML energy consumption** via **Intel RAPL**; used **band-pass filtering** and **K-means clustering** for noise reduction and layer segmentation; applied **polynomial regression** for **activation function energy profiling**; classified layer types via **Decision Tree** and **MLP ensemble**

PUBLICATIONS

7 papers (6 first-author) at IEEE IPDPS, IEEE ICNC, IEEE CCNC, ACM TOIT, and Springer SecureComm; 2 under review at ACM TOPS.
Full list: johnny-dai-git.github.io/portfolio

PATENTS

Oven Temperature Control Method, Device and Computer-Readable Storage Medium — Granted [CN 108594898 B](#) | Dai Yuanjun et al.